Result 1 (determinant with named dimension). [mathlib, Matrix.det_mul]
Let n be a natural number. Let $m = n$. Let A be a matrix of size $n \times n$ over $\mathbb{C}$. Let B be a matrix of size $m \times m$ over $\mathbb{C}$. Then

$$\det(A\,B) = \det(A)\ \det(B).$$

Result 2 (transpose with named dimension). [mathlib, Matrix.transpose_transpose]
Let n be a natural number. Let $m = n$. Let A be a matrix of size $m \times n$ over $\mathbb{C}$. Then

$$A^{\mathsf{T}\mathsf{T}} = A.$$

Result 3 (explicit dimension literal). [Dimension interfaces, vector identity]
Let x be a vector of length 3 over $\mathbb{R}$. Then

$$x = x.$$

Result 4 (dimension from record field). [Dimension interfaces, vector identity on stored dimension]
Let S be an element of SizedVector. Then

$$S.\text{vector} = S.\text{vector}.$$

LLM-assisted examples and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.